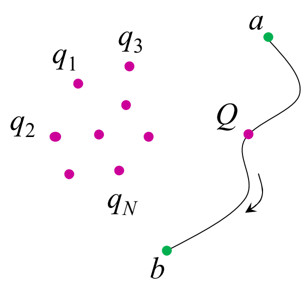
Work and Energy in Electrostatics

The work done to move a charge



$$\vec{\mathbf{F}}_{\text{system of charge}} = Q\vec{\mathbf{E}}$$

$$\vec{\mathbf{F}}_{\text{external agent}} = -Q\vec{\mathbf{E}}$$

$$W = \int_{\mathbf{a}}^{\mathbf{b}} \vec{\mathbf{F}} \cdot d\vec{\mathbf{l}} = -Q\int_{\mathbf{a}}^{\mathbf{b}} \vec{\mathbf{E}} \cdot d\vec{\mathbf{l}}$$

$$= -Q\,[V(\mathbf{b}) - V(\mathbf{a})]$$

independent of the path

electrostatic force is conservative

$$V(\mathbf{b}) - V(\mathbf{a}) = \frac{W}{Q}$$

zero

$$V(\vec{\mathbf{r}}) - V(\infty) = \frac{W}{Q}$$

$$W = QV(\vec{\mathbf{r}})$$

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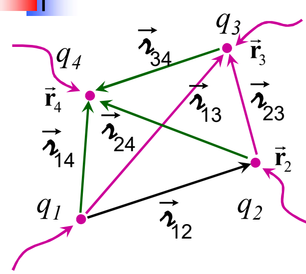
Electrostatic Energy for a Point Charge System



$$W_1 = 0$$

$$W_2 = q_2 V_1(\vec{\mathbf{r}}_2) = \frac{1}{4\pi\varepsilon_0} q_2\left(\frac{q_1}{\mathscr{r}_{12}}\right)$$

$$W_3 = q_3 V_{1,2}(\vec{\mathbf{r}}_3)$$

$$= \frac{1}{4\pi\varepsilon_0} q_3\left(\frac{q_1}{\mathscr{r}_{13}} + \frac{q_2}{\mathscr{r}_{23}}\right)$$

$$W_4 = \frac{1}{4\pi\varepsilon_0} q_4\left(\frac{q_1}{\mathscr{r}_{14}} + \frac{q_2}{\mathscr{r}_{24}} + \frac{q_3}{\mathscr{r}_{34}}\right)$$

$$W = W_1 + W_2 + W_3 + W_4$$

$$= \frac{1}{4\pi\varepsilon_0}\left(\frac{q_1q_2}{\mathscr{r}_{12}} + \frac{q_1q_3}{\mathscr{r}_{13}} + \frac{q_1q_4}{\mathscr{r}_{14}} + \frac{q_2q_3}{\mathscr{r}_{23}} + \frac{q_2q_4}{\mathscr{r}_{24}} + \frac{q_3q_4}{\mathscr{r}_{34}}\right)$$

$$W = \frac{1}{4\pi\varepsilon_0}\sum_{i=1}^{n}\sum_{j=1;j>i}^{n}\frac{q_iq_j}{\mathscr{r}_{ij}}$$

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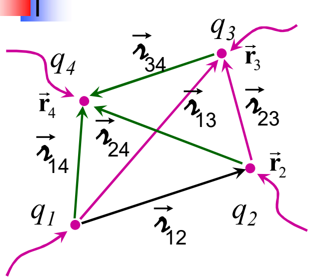
Electrostatic Energy for a Point Charge System



$$W = \frac{1}{4\pi\varepsilon_0}\sum_{i=1}^{n}\sum_{j=1;j>i}^{n}\frac{q_iq_j}{\mathscr{r}_{ij}}$$

$$W = \frac{1}{8\pi\varepsilon_0}\sum_{i=1}^{n}\sum_{j=1;j\neq i}^{n}\frac{q_iq_j}{\mathscr{r}_{ij}}$$

$$W = \frac{1}{2}\sum_{i=1}^{n} q_i\left(\sum_{j=1;j\neq i}^{n}\frac{1}{4\pi\varepsilon_0}\frac{q_j}{\mathscr{r}_{ij}}\right)$$

$$W = \frac{1}{2}\sum_{i=1}^{n} q_i V(\vec{\mathbf{r}}_i) \quad \text{Joule}$$

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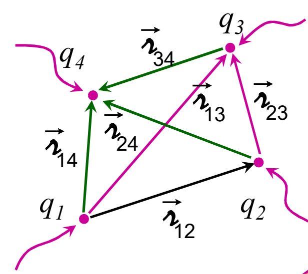
Electrostatic Energy of a Continuous Charge Distribution

$$\int_L \lambda V dl \qquad \int_S \sigma V da$$



$$W = \frac{1}{2}\sum_{i=1}^{n} q_i V(\vec{\mathbf{r}}_i) \quad \text{Joule}$$

$$W = \frac{1}{2}\int_V \rho V d\tau = \frac{\varepsilon_0}{2}\int_V (\vec{\nabla}\cdot\vec{\mathbf{E}})V d\tau$$

$$= \frac{\varepsilon_0}{2}\left[-\int_V \vec{\mathbf{E}}\cdot(\vec{\nabla}V)d\tau + \oint_S \vec{\mathbf{E}}V\cdot d\vec{\mathbf{a}}\right]$$

$$= \frac{\varepsilon_0}{2}\left(\int_V E^2 d\tau + \oint_S V\vec{\mathbf{E}}\cdot d\vec{\mathbf{a}}\right)$$

Using $\vec{\mathbf{E}} = -\vec{\nabla}V$ and $\rho = \varepsilon_0\vec{\nabla}\cdot\vec{\mathbf{E}}$

Electrostatic Energy Density:

$$= \frac{1}{2}\rho V = \frac{1}{2}\varepsilon_0 E^2 \quad \text{J/m}^3$$

$$\boxed{W = \frac{\varepsilon_0}{2}\int_{\text{all space}} E^2 d\tau} \quad \text{Joule}$$

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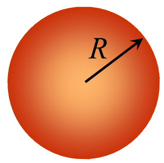
Electrostatic Energy of a Continuous Charge Distribution: Example

Energy stored in a Uniformly charged solid sphere of radius R & charge q.

$$W = \frac{1}{2}\int_V \rho V d\tau$$



Volume of integration

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Electrostatic Energy of a Continuous Charge Distribution: Example

Energy stored in a Uniformly charged solid sphere of radius R & charge q.

$$W = \frac{1}{2}\int_V \rho V d\tau$$

$$V(\vec{r}) = -\int_\infty^{\vec{r}} \vec{E}\cdot d\vec{l}$$

outside the sphere — inside the sphere

$$V(\vec{r}) = -\int_\infty^R \frac{1}{4\pi\varepsilon_0}\frac{q}{r^2}\hat{r}\cdot dr\,\hat{r} - \int_R^r \frac{1}{4\pi\varepsilon_0}\frac{qr}{R^3}\hat{r}\cdot dr\,\hat{r}$$

$$= \frac{1}{4\pi\varepsilon_0}\frac{q}{2R}\left(3-\frac{r^2}{R^2}\right)$$

Volume of integration

$$W = \frac{1}{2}\rho\frac{1}{\cancel{4\pi}\varepsilon_0}\frac{q}{2R}\int_0^R\left(3-\frac{r^2}{R^2}\right)\cancel{4\pi} r^2 dr = \frac{\rho q}{4\varepsilon_0 R}\left(3\frac{r^3}{3}-\frac{1}{R^2}\frac{r^5}{5}\right)\Bigg|_0^R = \frac{\rho q}{5\varepsilon_0}R^2$$

$$= \frac{q}{\frac{4}{3}\pi R^{\cancel{3}}}\frac{q}{5\varepsilon_0}\cancel{R^2} = \boxed{\frac{1}{4\pi\varepsilon_0}\frac{3}{5}\frac{q^2}{R}}$$

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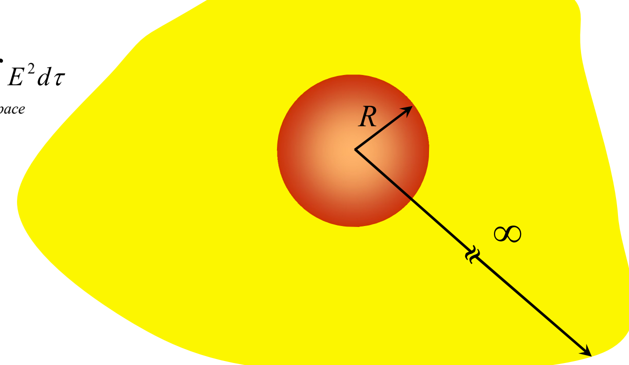
Electrostatic Energy of a Continuous Charge Distribution: Example

Energy stored in a Uniformly charged solid sphere of radius R & charge q.

$$W = \frac{\varepsilon_0}{2}\int_{all\ space} E^2 d\tau$$



Volume of integration: Entire Space

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Electrostatic Energy of a Continuous Charge Distribution: Example

Energy stored in a Uniformly charged solid sphere of radius R & charge q.

$$W = \frac{\varepsilon_0}{2}\int_{all\ space} E^2 d\tau$$

$$\vec{E} = \frac{1}{4\pi\varepsilon_0}\frac{q}{r^2}\hat{r} \qquad r > R$$

$$\boxed{\int_0^{\pi}\int_0^{2\pi}\sin\theta\, d\theta\, d\varphi = 4\pi}$$

$$\vec{E} = \frac{1}{4\pi\varepsilon_0}\frac{q}{R^3}r\,\hat{r} \qquad r < R$$

$$W = \frac{\varepsilon_0}{2}\left(\frac{q}{4\pi\varepsilon_0}\right)^2\left\{\int_0^R\int_0^{\pi}\int_0^{2\pi}\left(\frac{r}{R^3}\right)^2 r^2\sin\theta\, dr\, d\theta\, d\varphi + \int_R^{\infty}\int_0^{\pi}\int_0^{2\pi}\frac{1}{r^4}r^2\sin\theta\, dr\, d\theta\, d\varphi\right\}$$

$$= \frac{\varepsilon_0}{2}\left(\frac{q}{4\pi\varepsilon_0}\right)^2\left\{4\pi\int_0^R\left(\frac{r}{R^3}\right)^2 r^2 dr + 4\pi\int_R^{\infty}\frac{1}{r^2}dr\right\} = \frac{1}{4\pi\varepsilon_0}\frac{q^2}{2}\left(\frac{1}{5R}+\frac{1}{R}\right)$$

$$= \boxed{\frac{1}{4\pi\varepsilon_0}\frac{3}{5}\frac{q^2}{R}}$$

Volume of integration: Entire Space

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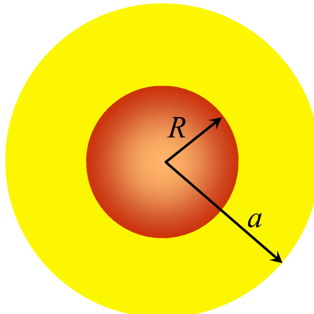
Electrostatic Energy of a Continuous Charge Distribution: Example

Energy stored in a Uniformly charged solid sphere of radius R & charge q.

$$W = \frac{\varepsilon_0}{2}\left(\int_V E^2 d\tau + \oint_S V\vec{E}\cdot d\vec{a}\right)$$



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Electrostatic Energy of a Continuous Charge Distribution: Example

Energy stored in a Uniformly charged solid sphere of radius R & charge q.

$$W = \frac{\varepsilon_0}{2}\left(\int_V E^2 d\tau + \oint_S V\vec{E}\cdot d\vec{a}\right)$$

$$\int_V = \frac{\varepsilon_0}{2}\left(\frac{q}{4\pi\varepsilon_0}\right)^2\left\{\int_0^R\int_0^\pi\int_0^{2\pi}\left(\frac{r}{R^3}\right)^2 r^2\sin\theta\,dr\,d\theta\,d\phi + \int_R^a\int_0^\pi\int_0^{2\pi}\frac{1}{r^4}r^2\sin\theta\,dr\,d\theta\,d\phi\right\}$$

$$= \frac{\varepsilon_0}{2}\left(\frac{q}{4\pi\varepsilon_0}\right)^2\left\{4\pi\int_0^R\left(\frac{r}{R^3}\right)^2 r^2 dr + 4\pi\int_R^a\frac{1}{r^4}r^2 dr\right\} = \frac{1}{4\pi\varepsilon_0}\frac{q^2}{2}\left(\frac{1}{5R}-\frac{1}{a}+\frac{1}{R}\right)$$

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Electrostatic Energy of a Continuous Charge Distribution: Example

Energy stored in a Uniformly charged solid sphere of radius R & charge q.

$$W = \frac{\varepsilon_0}{2}\left(\int_V E^2 d\tau + \oint_S V\vec{E}\cdot d\vec{a}\right)$$

$$\int_S = \frac{\varepsilon_0}{2}\int_0^{2\pi}\int_0^{\pi}\left(\frac{1}{4\pi\varepsilon_0}\frac{q}{a}\right)\left(\frac{1}{4\pi\varepsilon_0}\frac{q}{a^2}\right)a^2\sin\theta\,d\theta\,d\phi$$

$$= \frac{\varepsilon_0}{2}\left(\frac{q}{4\pi\varepsilon_0}\right)^2\frac{1}{a}4\pi = \frac{1}{4\pi\varepsilon_0}\frac{q^2}{2}\frac{1}{a}$$

$$\int_V + \int_S = \frac{1}{4\pi\varepsilon_0}\frac{q^2}{2}\left(\frac{1}{5R}-\cancel{\frac{1}{a}}+\frac{1}{R}\right) + \frac{1}{4\pi\varepsilon_0}\frac{q^2}{2}\cancel{\frac{1}{a}} = \boxed{\frac{1}{4\pi\varepsilon_0}\frac{3}{5}\frac{q^2}{R}}$$

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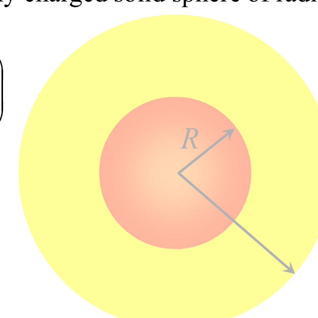
Electrostatic Energy of a Continuous Charge Distribution: Example

Energy stored in a Uniformly charged solid sphere of radius R & charge q.

$$W = \frac{\varepsilon_0}{2}\left(\int_V E^2 d\tau + \oint_S V\vec{E}\cdot d\vec{a}\right)$$



$$\int_V = \frac{1}{4\pi\varepsilon_0}\frac{q^2}{2}\left(\frac{1}{5R}-\frac{1}{a}+\frac{1}{R}\right)$$

$$\int_S = \frac{1}{4\pi\varepsilon_0}\frac{q^2}{2}\frac{1}{a}$$

As $a\to\infty$ $\quad \int_V = \frac{1}{4\pi\varepsilon_0}\frac{q^2}{2}\left(\frac{1}{5R}-\cancel{\frac{1}{a}}+\frac{1}{R}\right)$

As $a\to\infty$ $\quad \int_S = 0$

$$\boxed{W = \frac{1}{4\pi\varepsilon_0}\frac{3}{5}\frac{q^2}{R}}$$

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Electrostatic Energy of a Continuous Charge Distribution: Example

Home Work: Example 2.8 (pp 94)

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Electrostatic Energy: Paradox

$W = \frac{1}{2}\sum_{i=1}^{n} q_i V(\vec{r}_i)$ Energy stored in a stationary charge system can be +/–

$W = \frac{\varepsilon_0}{2}\int_{all\ space} E^2 d\tau$ Integrand being +ve definite, energy can never be –ve!

What has gone wrong?

To answer this, we shall first calculate the energy of a point charge which is the work to be done if the charge q would be condensed from an infinitely distributed charge cloud to a point in space

$$W = \frac{\varepsilon_0}{2}\left(\frac{q}{4\pi\varepsilon_0}\right)^2 \int_{r=\infty}^{0}\int_0^{\pi}\int_0^{2\pi}\left(\frac{1}{r^2}\right)^2 r^2 \underline{\sin\theta}\, dr\, \underline{d\theta\, d\phi}$$

$$= \frac{\varepsilon_0}{2}\left(\frac{q}{4\pi\varepsilon_0}\right)^2 4\pi \int_{r=\infty}^{0}\frac{1}{r^2}dr = \infty$$

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Electrostatic Energy: Paradox

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$W = \frac{\varepsilon_0}{2}\int_{all\ space} E^2 d\tau$ Integrand being +ve definite, energy can never be –ve!

What has gone wrong?

Answer: Nothing. We need to understand the generalization from point charge system to continuous charge system carefully

- we excluded the identical indices terms for point charge system.
- means, we excluded the contribution from q_i at $V(\vec{r}_i)$, i.e. the self energy of a point charge (which is infinite!)
- This exclusion is not evident in generalization as we have to integrate over entire space. Hence, doing so, we include the self energy of a continuous charge distribution which is very large.

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Electrostatic Energy: Paradox

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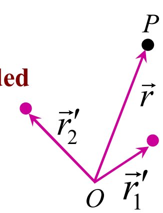
$W = \frac{\varepsilon_0}{2}\int_{all\ space} E^2 d\tau$ Integrand being +ve definite, energy can never be –ve!

What has gone wrong?

Lets consider an example: calculate total energy needed to assemble two point charges



$$W = \frac{1}{2}\sum_{i=1}^{n} q_i V(\vec{r}_i)$$

$$W = \frac{1}{2}\left(q_1 V(\vec{r}_1) + q_2 V(\vec{r}_2)\right) = \frac{1}{2}\frac{1}{4\pi\varepsilon_0}\left(q_1\frac{q_2}{\mathscr{r}_{21}} + q_2\frac{q_1}{\mathscr{r}_{12}}\right)$$

$$= \frac{1}{4\pi\varepsilon_0}\frac{q_1 q_2}{\mathscr{r}_{12}}$$

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Electrostatic Energy: Paradox

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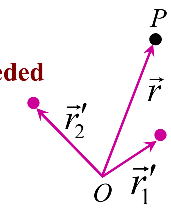

$$\vec{E}_1(\vec{r}) = \frac{1}{4\pi\varepsilon_0}\frac{q_1(\vec{r}-\vec{r}_1)}{|\vec{r}-\vec{r}_1|^3} \qquad \vec{E}_2(\vec{r}) = \frac{1}{4\pi\varepsilon_0}\frac{q_2(\vec{r}-\vec{r}_2)}{|\vec{r}-\vec{r}_2|^3}$$

$$\vec{E} = \vec{E}_1 + \vec{E}_2 \qquad W = \frac{\varepsilon_0}{2}\int_{all\ space}\left(E_1^2 + E_2^2 + 2\vec{E}_1\cdot\vec{E}_2\right)d\tau = W_1 + W_2 + W_{12}$$

$$W_{12} = \frac{\varepsilon_0}{2}\left(\frac{1}{4\pi\varepsilon_0}\right)^2 \int_{all\ space}\frac{q_1 q_2(\vec{r}-\vec{r}_1)\cdot(\vec{r}-\vec{r}_2)}{|\vec{r}-\vec{r}_1|^3|\vec{r}-\vec{r}_2|^3}d\tau$$

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Electrostatic Energy: Paradox

$W = \frac{1}{2}\sum_{i=1}^{n} q_i V(\vec{r}_i)$ Energy stored in a stationary charge system can be +/–

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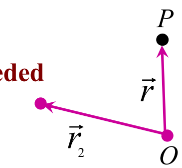
What has gone wrong?

Lets consider an example: calculate total energy needed to assemble two point charges



$$\vec{E}_1(\vec{r}) = \frac{1}{4\pi\varepsilon_0}\frac{q_1(\vec{r}-\vec{r}_1)}{|\vec{r}-\vec{r}_1|^3} \qquad \vec{E}_2(\vec{r}) = \frac{1}{4\pi\varepsilon_0}\frac{q_2(\vec{r}-\vec{r}_2)}{|\vec{r}-\vec{r}_2|^3}$$

$$\vec{E} = \vec{E}_1 + \vec{E}_2 \qquad W = \frac{\varepsilon_0}{2}\int_{all\ space}\left(E_1^2 + E_2^2 + 2\vec{E}_1\cdot\vec{E}_2\right)d\tau = W_1 + W_2 + W_{12}$$

$$W_{12} = \frac{\varepsilon_0}{2}\left(\frac{1}{4\pi\varepsilon_0}\right)^2 \int_{all\ space}\frac{q_1 q_2(\vec{r}-\vec{r}_1)\cdot(\vec{r}-\vec{r}_2)}{|\vec{r}-\vec{r}_1|^3|\vec{r}-\vec{r}_2|^3}d\tau$$

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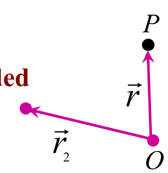
Electrostatic Energy: Paradox

$W = \frac{1}{2}\sum_{i=1}^{n} q_i V(\vec{r}_i)$ Energy stored in a stationary charge system can be +/–

$W = \frac{\varepsilon_0}{2}\int_{all\ space} E^2 d\tau$ Integrand being +ve definite, energy can never be –ve!

What has gone wrong?

Lets consider an example: calculate total energy needed to assemble two point charges



$$\vec{E}_1(\vec{r}) = \frac{1}{4\pi\varepsilon_0}\frac{q_1(\vec{r}-\vec{r}_1)}{|\vec{r}-\vec{r}_1|^3} \qquad \vec{E}_2(\vec{r}) = \frac{1}{4\pi\varepsilon_0}\frac{q_2(\vec{r}-\vec{r}_2)}{|\vec{r}-\vec{r}_2|^3}$$

$$\vec{E} = \vec{E}_1 + \vec{E}_2 \qquad W = \frac{\varepsilon_0}{2}\int_{all\ space}\left(E_1^2 + E_2^2 + 2\vec{E}_1\cdot\vec{E}_2\right)d\tau = W_1 + W_2 + W_{12}$$

$$W_{12} = \frac{\varepsilon_0}{2}\left(\frac{1}{4\pi\varepsilon_0}\right)^2 q_1 q_2 \int_{all\ space}\frac{\vec{r}\cdot(\vec{r}-\vec{r}_2)}{r^3|\vec{r}-\vec{r}_2|^3}d\tau = \frac{1}{4\pi\varepsilon_0}\frac{q_1 q_2}{r_2}$$

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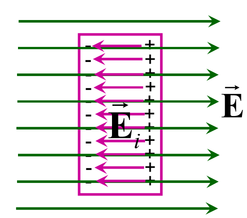
Conductors in Electrostatics

- In *Electrostatic* constraint, **E** = 0 inside a conductor.

  This is true even if conductor is charged or placed in an external electric field.



Free charges will continue to flow until the field cancellation within the conductor is complete.

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Conductors in Electrostatics

- From the Gauss's Law, $\vec{\nabla}\cdot\vec{\mathbf{E}}=\rho/\varepsilon_o$ and as $\vec{\mathbf{E}}=0$ inside a conductor, $\rho=0$ inside the conductor.
- Net charge resides completely on the surface (minimum energy configuration).

  Energy of a sphere if the charges are on the surface: $\frac{1}{8\pi\varepsilon_0}\frac{q^2}{R}$ 0.125
  (*Example:* 2.8)

  Energy of a sphere if the charges are over the volume: $\frac{3}{20\pi\varepsilon_0}\frac{q^2}{R}$ 0.15
  (*Problem:* 2.32)
- Consider any two points a and b on/within the conductor:

  $V(\vec{\mathbf{b}})-V(\vec{\mathbf{a}})=-\int_{\vec{\mathbf{a}}}^{\vec{\mathbf{b}}}\vec{\mathbf{E}}\cdot d\vec{\mathbf{l}}=0$ $\longrightarrow$ Conductor, including its surface, is an equipotential

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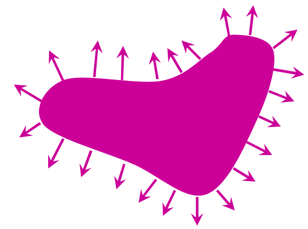
Conductors in Electrostatics

- Electric field is every where normal to the conductor's surface



as the surface is equipotential, gradient of V has to be normal to it.

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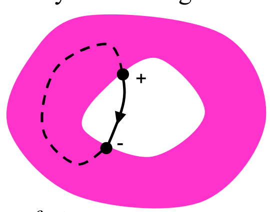
Conductors with an empty cavity

- If cavity (surrounded by conducting material) is empty of charges, field within the cavity is zero.
- As the field within the meat of the conductor is zero, for an empty cavity, field line will begin and end on the cavity wall owing to the charges on the wall.

  Let us compute the line integral of the electric field around a closed path, part of which is along the field line inside the cavity, and part in the conductor.



$$\oint\vec{\mathbf{E}}\cdot d\vec{\mathbf{l}}=\int_{dash}\vec{\mathbf{E}}\cdot d\vec{\mathbf{l}}+\int_{cont}\vec{\mathbf{E}}\cdot d\vec{\mathbf{l}}=0+\int_{cont}\vec{\mathbf{E}}\cdot d\vec{\mathbf{l}}=finite$$

but $\oint\vec{\mathbf{E}}\cdot d\vec{\mathbf{l}}=0$ $\longrightarrow$ Field within an empty cavity is zero and there is no charge on the cavity wall.

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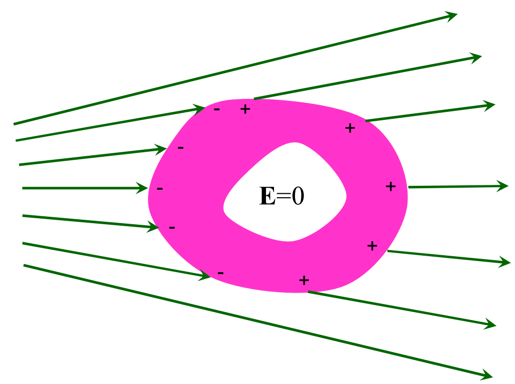
Conductors with an empty cavity: Faraday Cage



Space inside a cavity is safely shielded from the external fields!!

No external field penetrates the conductor!

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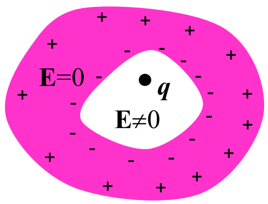
Conductors with a cavity having charge in it



No external field penetrates the conductor!

But the outside world would know about the presence of the charge within the cavity because of the induced charge at the external surface of the conductor

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Poisson's and Laplace Equations

$$\vec{\nabla}\cdot\vec{\mathbf{E}} = \frac{\rho}{\varepsilon_0} \qquad \vec{\nabla}\times\vec{\mathbf{E}} = 0 \qquad \vec{\mathbf{E}} = -\vec{\nabla}V$$

$$\vec{\nabla}\cdot\vec{\mathbf{E}} = \vec{\nabla}\cdot\left(-\vec{\nabla}V\right) = -\nabla^2 V$$

$$\boxed{\nabla^2 V = -\frac{\rho}{\varepsilon_0}}$$ Poisson's Equation

In the region, where there is no charge,

$$\boxed{\nabla^2 V = 0}$$ Laplace's Equation

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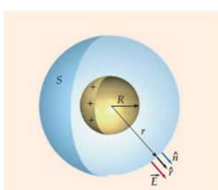
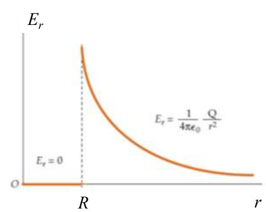
Electrostatic Boundary Conditions

Electric field due to thin spherical shell of charge

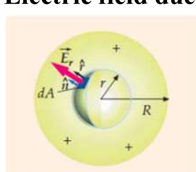


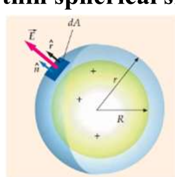
$$\vec{\mathbf{E}}_r = \frac{1}{4\pi\varepsilon_0}\frac{Q}{r^2} \qquad r < R$$

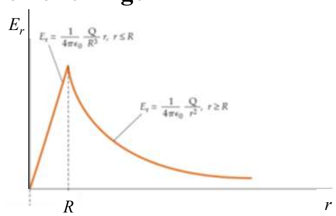
$$\vec{\mathbf{E}}_r = 0 \qquad r < R$$



Electric field due to thin spherical shell of charge







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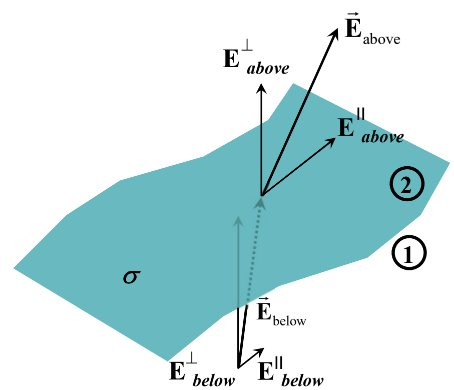
Electrostatic Boundary Conditions

Electric field suffers discontinuity at the surface charge



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Electrostatic Boundary Conditions

Electric field suffers discontinuity at the surface charge

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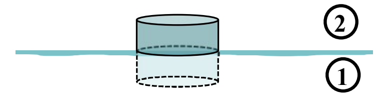
Electrostatic Boundary Conditions

Electric field suffers discontinuity at the surface charge



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Electrostatic Boundary Conditions

Electric field suffers discontinuity at the surface charge

$$\vec{\nabla}\cdot\vec{\mathbf{E}} = \rho/\varepsilon_0$$

$$\int_V (\vec{\nabla}\cdot\vec{\mathbf{E}})\,d\tau = \oint \vec{\mathbf{E}}\cdot d\vec{a} = \vec{\mathbf{E}}_2\cdot\hat{n}_2\Delta a + \vec{\mathbf{E}}_1\cdot\hat{n}_1\Delta a + w$$

$$= \vec{\mathbf{E}}_2\cdot\hat{n}\Delta a - \vec{\mathbf{E}}_1\cdot\hat{n}\Delta a + w = \hat{n}\cdot(\vec{\mathbf{E}}_2-\vec{\mathbf{E}}_1)\Delta a + w$$

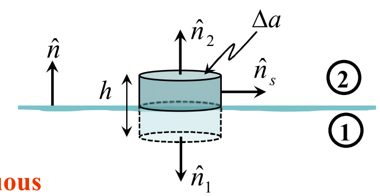
$$\lim_{h\to 0}\hat{n}\cdot(\vec{\mathbf{E}}_2-\vec{\mathbf{E}}_1)\cancel{\Delta a} + \cancel{\lim_{h\to 0} w} = \lim_{h\to 0}\int_V \frac{\rho}{\varepsilon_0}\,d\tau = \lim_{h\to 0}\frac{\rho}{\varepsilon_0}h\cancel{\Delta a}$$

Total charge Δq through the volume $h\Delta a$:

$$\Delta q = \rho\, h\, \Delta a \qquad \lim_{h\to 0}\Delta q = \lim_{h\to 0}\left(\rho\, h\, \Delta a\right) = \sigma\,\Delta a$$

$$\boxed{\hat{n}\cdot(\vec{\mathbf{E}}_2-\vec{\mathbf{E}}_1) = \sigma/\varepsilon_0}$$

$$E_{2n} - E_{1n} = \sigma/\varepsilon_0$$



Normal comps of E are discontinuous

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Electrostatic Boundary Conditions

Electric field suffers discontinuity at the surface charge

$$\vec{\nabla}\times\vec{\mathbf{E}} = 0 \;\Rightarrow\; \int_S (\vec{\nabla}\times\vec{\mathbf{E}})\cdot d\vec{a} = 0 \;\Rightarrow\; \oint_C \vec{\mathbf{E}}\cdot d\vec{l} = 0$$

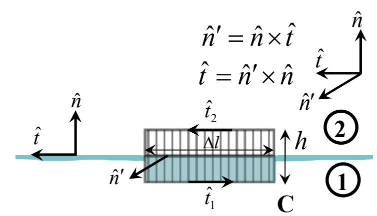
$$\vec{\mathbf{E}}_2\cdot\hat{t}_2\Delta l + \vec{\mathbf{E}}_1\cdot\hat{t}_1\Delta l + w = \vec{\mathbf{E}}_2\cdot\hat{t}\Delta l - \vec{\mathbf{E}}_1\cdot\hat{t}\Delta l + w = \hat{t}\cdot(\vec{\mathbf{E}}_2-\vec{\mathbf{E}}_1)\Delta l + w = 0$$

$$= (\hat{n}'\times\hat{n})\cdot(\vec{\mathbf{E}}_2-\vec{\mathbf{E}}_1)\Delta l + w = \hat{n}'\cdot\hat{n}\times(\vec{\mathbf{E}}_2-\vec{\mathbf{E}}_1)\Delta l + w = 0$$

$$\lim_{h\to 0}\left(\hat{n}'\cdot\hat{n}\times(\vec{\mathbf{E}}_2-\vec{\mathbf{E}}_1)\Delta l + \cancel{w}\right) = 0$$

$$\hat{n}'\cdot\left[\hat{n}\times(\vec{\mathbf{E}}_2-\vec{\mathbf{E}}_1)\right] = 0$$

$$\boxed{\hat{n}\times(\vec{\mathbf{E}}_2-\vec{\mathbf{E}}_1) = 0}$$



Tangential comps of E are continuous

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Electrostatic Boundary Conditions

Electric field suffers discontinuity at the surface charge

$$\vec{\nabla}\times\vec{\mathbf{E}}=0 \Rightarrow \boxed{\hat{n}\times(\vec{\mathbf{E}}_2-\vec{\mathbf{E}}_1)=0}$$

$$\boxed{\hat{n}\times(\vec{\mathbf{E}}_2^{\parallel}-\vec{\mathbf{E}}_1^{\parallel})=0}$$

$$\boxed{\vec{\mathbf{E}}_2^{\parallel}-\vec{\mathbf{E}}_1^{\parallel}=0}$$

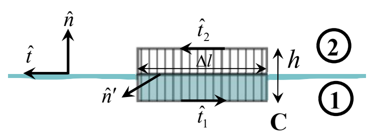
$$\vec{\mathbf{E}}=\vec{\mathbf{E}}^{\perp}+\vec{\mathbf{E}}^{\parallel}=E^{\perp}\hat{n}+\vec{\mathbf{E}}^{\parallel}$$

$$\hat{n}\times\vec{\mathbf{E}}=\hat{n}\times\left(E^{\perp}\hat{n}+\vec{\mathbf{E}}^{\parallel}\right)=\hat{n}\times\vec{\mathbf{E}}^{\parallel}$$

$$(\hat{n}\times\vec{\mathbf{E}})\times\hat{n}=-\hat{n}\times(\hat{n}\times\vec{\mathbf{E}})$$

$$=-[\hat{n}(\hat{n}\cdot\vec{\mathbf{E}})-\vec{\mathbf{E}}(\hat{n}\cdot\hat{n})]$$

$$=\vec{\mathbf{E}}-E^{\perp}\hat{n}=\vec{\mathbf{E}}^{\parallel}$$



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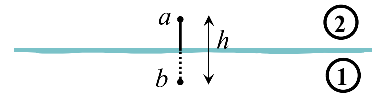
Electrostatic Boundary Conditions

Electric potential



$$V_{above}-V_{below}=-\int_a^b\vec{\mathbf{E}}\cdot d\vec{l}$$

$$\lim_{dl\to 0}\int_a^b\vec{\mathbf{E}}\cdot d\vec{l}=0$$

$$\boxed{V_{above}=V_{below}}$$

$$\hat{n}\cdot(\vec{\mathbf{E}}_2-\vec{\mathbf{E}}_1)=\sigma/\varepsilon_0 \qquad \vec{\mathbf{E}}=-\vec{\nabla}V$$

$$\Rightarrow \hat{n}\cdot(\vec{\nabla}V_{\text{above}}-\vec{\nabla}V_{\text{below}})=-\sigma/\varepsilon_0$$

$$\Rightarrow \boxed{\frac{\partial V_{\text{above}}}{\partial n}-\frac{\partial V_{\text{below}}}{\partial n}=-\frac{\sigma}{\varepsilon_0}} \qquad \because \frac{\partial V}{\partial n}=\vec{\nabla}V\cdot\hat{n}$$

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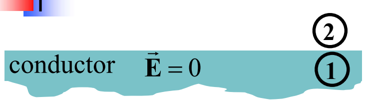
Surface charge and force on a Conductor



$$\vec{\mathbf{E}}_{above}-\vec{\mathbf{E}}_{below}=\frac{\sigma}{\varepsilon_0}\hat{n}$$

$$\vec{\mathbf{E}}=\frac{\sigma}{\varepsilon_0}\hat{n} \Rightarrow \sigma=-\varepsilon_0\frac{\partial V}{\partial n}$$

In the presence of an electric field, a surface charge will experience a force. Force per unit area $\vec{f}=\sigma\vec{\mathbf{E}}$ $\quad(\vec{f}=q\vec{\mathbf{E}})$

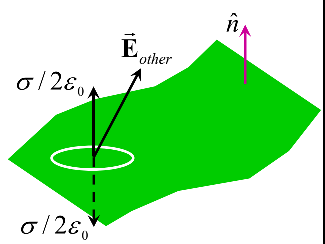
$$\boxed{\vec{f}=\frac{1}{2\varepsilon_0}\sigma^2\hat{n}}$$

$$\vec{f}=\sigma\vec{\mathbf{E}}_{ave}=\frac{1}{2}(\vec{\mathbf{E}}_{above}+\vec{\mathbf{E}}_{below}) \qquad \vec{\mathbf{E}}=\cancel{\vec{\mathbf{E}}_{patch}}+\vec{\mathbf{E}}_{other}$$

$$\vec{\mathbf{E}}_{above}=\vec{\mathbf{E}}_{other}+\frac{\sigma}{2\varepsilon_0}\hat{n}$$

$$\vec{\mathbf{E}}_{below}=\vec{\mathbf{E}}_{other}-\frac{\sigma}{2\varepsilon_0}\hat{n}$$

$$\vec{\mathbf{E}}_{other}=\frac{1}{2}(\vec{\mathbf{E}}_{above}+\vec{\mathbf{E}}_{below})=\vec{\mathbf{E}}_{ave}=\frac{\sigma}{2\varepsilon_0}\hat{n}$$



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